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Extending Neurophenomenology

Connecting First- and Third-Person Perspectives through Their Timescales

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There is a seemingly insurmountable gap between the neuronal states of the brain and the phenomenal states of consciousness. The various solutions that have been developed to solve this dilemma have been unsatisfactory because, as we argue, of the lack of shared features serving as "common currency" or as a bridge between neuronal and phenomenal states. There are significant differences between neuronal and phenomenal states, which, on the methodological side, correspond to third-person and first-person approaches, respectively. We propose a new neurophilosophical explanatory strategy to connect neuronal and phenomenal states by combining a first-person approach with a third-person approach through the integration of their distinct timescales, e.g., longer (first PP) and shorter (third PP). By virtue of their cross-timescale relationships, both perspectives are integrated with each other; that, in turn, allows us to take into view the direct temporal and spatial connection of neuronal and phenomenal states. Together, our epistemological approach allows to extend current neurophenomenology by providing direct time-scale based connection of first- and third-person perspective which connects neuronal and phenomenal states in a temporal way. This is suggested by the Temporospatial theory of consciousness (TTC), which methodologically presupposes a neurophenomenological approach through the temporal integration and convergence of first and third-person perspectives.
Keywords: Neurophilosophy, Consciousness, Neurophenomenology, Timescale, Spatiotemporal Connection.

1. Introduction

Consciousness is both an old and a novel problem. The reason for describing it with such a set of seemingly contradictory adjectives is, on the one hand, that thinking about consciousness was embedded in philosophical studies a long time ago, which we can even trace to some texts from the ancient Greek period. While on the other hand, the placing of consciousness in a scientific context is probably not more than a century

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old. The scientific study of consciousness in neuroscience relies on the basic assumption that the brain determines the mind (Uttar, 2011). In particular the development of neuroimaging techniques in neuroscience with respect to measurement and localization demonstrate the foundational role of neuronal activity in human mental phenomena like consciousness. While many philosophers have also acquiesced in this consensus, they have insisted that features of the mind such as consciousness have properties such as intentionality, subjectivity, and phenomenality that are completely heterogeneous from physical features like the brain, thus leading to an explanatory gap between neuronal and phenomenal states (Levine, 1983). This raises the question how to deal with the tension between the two, neuronal and phenomenal states which also touches upon the methodological strategy. One may, for instance, either pursue a reductionist or non-reductionist approach in resolving their explanatory gap and ultimately what also has been coined as "hard problem" (Chalmers, 1995).

The rupture between the neuronal states of the brain and the phenomenal states of consciousness has long been emphasized in philosophy. This problem has been referred to in more recent times as the mind-body problem, i.e., how exactly the mind and the body, which are regarded as strictly distinct, are united (Fodor, 1981). In Descartes' argument on the interaction between mind and body, he believed that humans consist of an extended body and a thinking mind, which are two distinct entities. He identified something called the pineal gland in the brain, which can provide information about the body to the brain via "vital essence" and issue commands from the mind to the body (Shapiro, 2011). Descartes' theory is a double-edged sword. The positive side is that his attempts emphasized the fact that the brain (including neuronal states) and the mind (including phenomenal states) are intimately connected. The negative side is that he fails to do so, completely severing the two in a dualistic fashion.

Methodologically, we are confronted with the challenge of connecting first-person concepts, e.g., phenomenal concepts like qualia, etc., with third-person facts like neural data. This leads us to the crossroads of philosophy and neuroscience. While philosophy focuses on concepts including first-person phenomenal concepts, science like neuroscience deals with data and facts, e.g., neural fact as obtained in third-person perspective. Connecting neural and phenomenal states thus confronts us with a methodological challenge. Going beyond both, we need to intimately connect their respective methodological strategies with each other in order to take into view the necessary connection of neural and phenomenal states. Based on our prior methodological work (Northoff, 2004; 2014b; 2022), we propose a new methodological strategy with a particular epistemological approach to the question of first- and third-person perspective. The combination of first- and third-person approaches is realized through the integration between the longer timescales of the first-person perspective and the shorter timescales of the third-person perspective. Thus, neuronal states are linked to phenomenal states in a temporal and spatial way which methodologically can be taken into view through the temporal integration and convergence of the temporal features of first- and third-person perspectives.

2. Consciousness: Neuronal States and Phenomenal States

2.1. Characteristics of Neuronal States

The neuron in the brain, also called a neuron cell, consists of three parts: the cytosol, dendrites, and axons, and is the smallest unit of structure and function of the nervous system. Its basic role and function is to receive and transmit information, and the connections established between neurons through synapses form extremely complex neural circuits. The neural circuit is the basic unit of information processing in the brain. A neuron has two basic states, resting and excited (Snyder and Raichle, 2012). The former means that it is in a dormant state with no discharge when it is not receiving external stimuli or information. The latter implies that the neuron is stimulated or information enters and thus generates firing activity. When the resting state transitions to an excited state, the potential difference between the intracellular and extracellular membranes changes thereby creating a nerve impulse. Nerve impulses link neurons through neurotransmitters and turn on the exchange and processing of information. Therefore, the composition and function of a neuron determines its basic state and the transformation of its state, which in turn is the beginning of brain and body activity. Neuroscientists usually place neuronal states in the cortical evolution and modular analysis of the brain to explore the secrets of human cognitive activity (Dunbar and Shultz, 2007). The neuronal state of the brain has three important characteristics.

First, objectivity. Objectivity or objective reality refers to the objective existence of things independent of us as investigators and our methodological bias. We can analyze the objective existence through empirical evidence and data without the influence of personal feelings, bias or subjective consciousness. Neurons have a specific physical form, such as the three parts of the cytosol, dendrites and axons. Neurons have a specific organizational structure, e.g., neurons build neural circuits through synapses. As well, neurons have specific states of existence, such as resting and excited states. Therefore, the neuronal state exhibits obvious objective characteristics.

Second, functionality. Functionality refers to the favorable role played by things or methods. To distinguish functionality from objectivity is to emphasize the possibility that functional properties and material content can be separated. Using the principle of the computer as a metaphor, it is not the material content as a component that determines how it operates, but rather the functional organization possessed by the system as a whole. The functionality of neurons in the brain involves a number of aspects such as information transmission, regulation, network structure and neuroplasticity (Aagaard, 2003). Together, these functions contribute to normal brain function and behavioral performance.

Third, mechanism. Mechanisms refer to the structural relationship and mode of operation between elements. The various elements of the neuron are linked together to form a stable cellular structure, and rely on action potential conduction and synaptic transmission to form the connection and communication between neurons. It is on the basis of the neuronal mechanism that the function of the neuron is properly exercised and the existence of the neuron is secured (Botvinick and Braver, 2015). The characterization of the mechanism is an important basis for the functioning of the organism and its vital activities.

2.2. Characteristics of Phenomenal States

The phenomenal state of consciousness is a unique property of consciousness, also known as phenomenal consciousness. It is often the source of the controversy into which the question of consciousness has fallen. Brock's classic account distinguishes between two kinds of consciousness, namely, access consciousness and phenomenal consciousness. A mental state is access consciousness when and only when it can be used as a premise for reasoning and naturally rationally controls action and speech. A mental state is phenomenal consciousness when and only when it is experienced from the first-person (Block, 1995). According to Block, access consciousness is a functional property of cognitive nature, while phenomenal consciousness is purely experiential, and the two are logically independent of each other. Our further analysis reveals that the phenomenal states of consciousness has three distinctive characteristics.

First, subjectivity or interiority. As Nagel puts it, we never seem to be able to know what it is like to experience anything other than what it is like to be a creature other than our self. For example what it is like to experience being a bat (Nagel, 1974). In other words, phenomenal states can only be presented from a first-person perspective.

Second, qualitative feature, the qualia (Block, 1981). It is someone's subjectively acquired experiential feelings such as pain, itch, redness, etc. If the subjective feature allows the subject to make a particular state conscious rather than non-conscious, then the qualitative feature is what makes the phenomenal state so this and not the other kind of phenomenal states.

Third, Intransitivity. Phenomenal consciousness is not like access consciousness, which is characterized by intentionality or transitivity. Access consciousness is a property about other things, always involving a content and pointing to an object (Brentano, 1995). In contrast, when we try to describe phenomenal consciousness, it does not involve a representation about an object, at least at the conceptual level.

To summarize, there is a significant difference between the neuronal features of the brain and the phenomenal features of consciousness. The former is objective, functional, and mechanistic, while the latter is subjective, qualitative, and intransitive. Both are important research elements in neurophilosophy, but how to study them with a unified set of methods has never been finalized. Thankfully, the search for similar shared features as a "common currency" between the two continues unabated (Northoff *et al*., 2020; 2025). We here focus on the possibility that both the brain and consciousness may possess spatiotemporal features. This work has been incorporated into what is known as Spatiotemporal neuroscience (Northoff *et al*., 2020; Northoff and Evers, 2021). Spatiotemporal neuroscience is an extension and complement to behaviorist and cognitivist neuroscience. Unlike in the latter two which determine the brain by either behavior or cognition, Spatiotemporal Neuroscience characterizes the brain by its intrinsic spatial and temporal features which, moreover, are supposed to provide direct connection of neuronal and mental states through their shared temporal and spatial features, e.g., their "common currency" (Northoff *et al*., 2020; 2025). This raises the question how we can take into view such shared features of neuronal and phenomenal states. That requires a particular methodological strategy that allows converging first and third-person perspectives, as through these we can access neuronal and phenomenal states respectively, that shall be the focus in the following.

3. Methodology of Consciousness Research: Third-Person Approach and First-Person Approach

3.1. Characteristics of the Third-Person Approach

The core of the third-person approach is observation (Ge *et al*., 2018). Observation is an outward-looking activity oriented towards the object-object, and is an important means and a necessary prerequisite for acquiring information, understanding phenomena and responding to them. It is often closely associated with the sense organs, including the use of sight, hearing, touch, smell and taste to perceive things and events in the surrounding environment. Advanced technology provides a variety of instruments to extend the scale of observation, thus allowing us to obtain more extensive and precise data. Modern science, with its "scientific materialism" (Whitehead, 1925) as its worldview, has focused heavily on the role of observation. It believes that the study of the brain as an object from a third-person perspective – including its structure, mode of operation, and function – is scientific and sufficient for consciousness. In neuroscience experiments, the third-person approach assesses the presence of consciousness by observing behavioral differences or stimulus responses in subjects. To further reveal the mechanisms of neuronal activity associated with consciousness, scientists used functional magnetic resonance imaging (fMRI), electroencephalography (EEG), and other neural signals to look for neural correlates of consciousness. They are trying to gather enough evidence to reveal which brain regions and neural activities are involved in consciousness and its content. In practice, this can at best explain so-called "easy problem". The phenomenal states that are the "hard problem" are completely ignored (Schneider and Velmans, 2017).

The third-person approach, with observation at the core of the methodology, has three characteristics.

First, it is quantitative. Quantitative analysis is the process of analyzing and interpreting data using numerical and statistical methods (Donahoe, 2002). In quantitative analysis, the researcher collects data, usually in numerical form, and then analyzes these data using mathematical and statistical techniques to understand the patterns, relationships, and trends of the phenomenon.

Second, it is publicly visible. Public visibility means that access to the same content by different subjects is reliable. The validity of the results can only be guaranteed if the public, and not just the private, is visible to the acquired data (Metag, 2021). In neuroscience, the study of the brain and its neurons is a public project intended to reveal features common to the human brain. This biological similarity spans gender, ethnicity and age. It is an important prerequisite for our research on the universality of consciousness.

Third, it is robust. Robustness, also known as reproducibility, refers to the inevitable occurrence of a result under specific conditions (Sun *et al.*, 2008). An experiment can only be generally accepted by the academic community if it can be repeated many times under the same conditions by independent researchers with similar or consistent results. Emphasis on robustness focuses on verifying the reliability of the results and eliminating the possibility of chance and error. By continually repeating experiments and observations, scientists can gain a deeper understanding of natural laws and phenomena, thereby advancing scientific knowledge.

3.2. Characteristics of the First-Person Approach

The core of the first-person approach is intuition (Symons, 2008). Intuition is an inward-looking activity oriented towards the subject itself and plays an important role in the philosophical tradition. It is an important way of integrating information, summarizing laws, and determining knowledge. Intuition emphasizes intellectual effort through reason, abstraction, or dialectics in order to set aside phenomena and thus grasp the essence of things. The first-person approach corroborates the phenomenal content of consciousness through intuitive insight and introspective training. It is based on the premise that the subject's experiential knowledge of the content of his or her consciousness is reliable, and requires the subject to report his or her mental state through language.

The greatest advantage of the first-person approach is the directness of the subject's report. It confronts the phenomenal state directly, rather than indirectly by approaching the neuronal states of the brain, thus leaving a "black box" between them. However, there are some problems with the first-person approach. The most important is the problem of "response preference" (Tune, 1964). Response preference refers to a subject's tendency to respond with uncertainty to a phenomenal state, such that the response thresholds set by different subjects for the same stimulus may vary depending on whether they are "conservativee" or "liberal", or a subject may change his or her initial response threshold because the validity of his or her report is rewarded or penalized. The blurring of subject thresholds results in a significant reduction in the authority and accuracy of subjective reports of the same phenomenal state.

The first-person approach, with intuition at the core of its methodology, has three characteristics.

First, it is qualitative (Cuthbertson *et al.*, 2020). The first-person approach can only assert the meaning and character of the object in question by describing, interpreting, and understanding it (Fossey *et al.*, 2002). And it cannot accurately calculate its variations and differences at finer granularities. After all, human thinking systems are fundamentally different from computers. A large portion of human cognitive processing of information is in an unconscious processing state, and is not realized by the subject until certain content reaches a fairly high threshold. This is unlike the computational process of a computer, where each and every arithmetic procedure is clearly quantifiable.

Second, it is private. Private, or privately owned, means that it can only be presented through the first-person channel (Rizzuto, 1993). The scope of validity of the first-person approach is strictly limited to the interior of the subject and is opaque to others. This is the source of the problem of other minds (Overgaard, 2006), whereby the individual has difficulty understanding or recognizing the inner feelings, intentions, or thoughts of others. Even though language can provide a good description of the subject's state and communicate between subjects, it still does not change the private nature of the first-person approach.

Third, it is sensitive. Sensitivity, also known as variability, means that the relationship between a result and the conditions of its realization is often uncertain, which makes it difficult to repeat. Because the first-person approach is directly confronted with the internal state of the subject, the extent to which it is affected by internal and external influences is uncertain. Again faced with a Gestalt image, while some will see a rabbit; some will see a duck's head, some will see a vase, while others will see two faces (Köhler, 1967).

4. Converging Method and Consciousness: From Perspectives to States

4.1. Neuronal States and the Third-Person Approach

Third-person approaches enable factual studies of neuronal states. Neuronal activity in the brain is generally measured or localized through brain imaging techniques to obtain experimental data related to consciousness. Measurement of consciousness is devoted to looking for neural markers in the brain's neural processing where consciousness occurs, such as P3 waves (Dehaene *et al.*, 2014), recurrent processing (Lamme and Roelfsema, 2000), re-entry (Edelman, 1993), and so on. Take, for example, the well-known neurobiological theory. Crick and Koch found experimentally that consciousness depends heavily on some form of short-term memory, and some form of continuous attention mechanism. It causes neurons to oscillate continuously in the range of 35-75 Hz, and in turn the brain acquires a temporary global unity. The ultimate conclusion is that consciousness is about neuronal oscillations of 35-75 Hz produced by the cerebral cortex (Francis and Koch, 1990). Theories based on the third-person approach commit to specific neuronal states as the key to the onset of consciousness. The localization of consciousness in the brain focuses on the definite location of consciousness in the brain posterior cortical regions are key in mediating consciousness (Storm *et al.*, 2017). This implies that different contents or functions of consciousness may correspond to different brain regions or neural networks. However, there are others who believe that the brain as a whole is involved in the process of consciousness (Northoff and Zilio, 2022a and b; Northoff and Huang, 2017; Zhang and Northoff, 2022). This is reflected in the localizationism *vs* holism debate about the locus of consciousness.

Let us further detail this with respect to the above-mentioned features of the third-person perspective.

First, the quantitative features of the third-person approach apply to neuronal states as objective beings. When a certain consciousness arises, the associated neuronal states of the brain can be shown to have computational features as measured by neuroimaging techniques. Such as functional relationships, graphical distributions, probability statistics, and so on.

Second, the communal character of the third-person approach is consistent with the functional role of neuronal states. The publicness feature is able to stably link specific brain regions or neuronal activity to specific functional properties. For example, through the study of publicly visible impairments, neuroscientists were able to discover that Broca's area is closely related to the ability to organize language (Jaredet *et al.*, 2010).

Finally, the robustness feature of the third-person approach facilitates the revelation of mechanisms inherent to neuronal activity. Some mechanists claim that a mechanism is a structure drawing on its constituent parts, its constituent operations, and their organization to perform a certain function. Mechanisms coordinate the performance of functions responsible for one or more phenomena (Flinker *et al.*, 2015). That is to say consciousness as a specific phenomenon comes from a specific neuronal mechanism. The first-person approach can identify the neuronal mechanisms responsible for consciousness through repeated attempts.

4.2. Phenomenal States and the First-Person Approach

The first-person approach enables the conceptual study of phenomenal states. The investigation of the content and structure of consciousness usually begins with reflection on the self. The first-person approach is a methodology in which phenomenology, introspection, and meditation are the primary tools. Because of the special nature of consciousness, it is the starting point of all human understanding. We cannot go beyond consciousness and rely on something else to measure it. This "cognitive primacy" (Thompson, 2014) gives the first-person perspective strong authority. That is, the belief that intuitively no other person can understand their internal experience better than the self. The phenomenological approach is based on the phenomenological tradition of reduction and suspension of natural attitudes (Pringle *et al.*, 2011). The nature of the subject's experience is understood through the steps of detailed description, in-depth analysis, and suspension of preconceptions. The introspective approach, on the other hand, relies on the ability to pay attention, scrutinize internal perceptions, and reflect them through reports (Jack and Shallice, 2001). Internal perceptions can come from specific stimuli or be purely meta-perceptual, which distinguishes between strong and weak introspective approaches. The approach of meditation derives from the meditative practices of the Buddhist and Vedic traditions (Sedlmeier *et al.*, 2012). Its claim is roughly that phenomenal states can be revealed in the self-revelation of consciousness. To illustrate with a metaphor, light can illuminate things around it, but light does not need other light to illuminate it. Thus, meditation practices are often combined with specific breathing, body postures, or attentional goals to help individuals enter a deep inner state. This state is calm and clear and facilitates a better understanding of our phenomenal states.

Thus, the first-person approach facilitates the study of phenomenal states.

First, the qualitative character of the first-person approach applies to phenomenal states as subjective beings. This is because phenomenal states of consciousness are difficult to deal with quantitatively. We can distinguish between pain and the absence of pain, or between the pain of a pinprick and the pain of a burn. But the distinction can only be made at a looser level than a meticulously delineated scale (Aydede, 2019). Similarly we can distinguish between red and green experiences, and we can distinguish between dark and light colored experiences (Levine, 2006). But the distinction can only be made on a vague scale, and cannot be given to experiences that correspond to any small change in color.

Second, the private character of the first-person approach fits the description of the felt quality of the phenomenal state. The first-person approach is confined within the subject; it has no possibility of reaching the other mind. And the perceptual quality of phenomenal states is likewise revealed only to the subject. Just as even though Mary learns all there is to know about the physical world, because she always lives in an environment where there are only two colors, black and white, she cannot have an experience of what red is like. Receptive qualities can only be felt privately (Jackson, 1986).

Finally, the sensitivity character of the first-person approach facilitates the determination of the intransitivity of phenomenal states. Because phenomenal states are not about other things, they often do not involve a definite object. The sensitivity of the first-person approach is more able to focus on the pure experience of the phenomenal state. For a phenomenal state such as the feeling of heat that I experience standing in the sun at moment t1, there is no guarantee that it will be felt either by another person standing in the sun at moment t1, or by myself standing in the sun at moment t2. This makes the third-person approach difficult to apply.

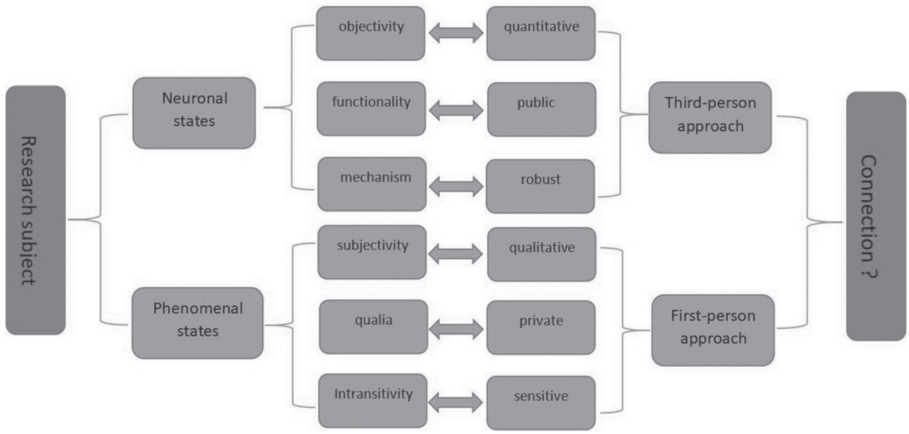

FIG. 1. Schematic illustration of the features of neuronal and phenomenal states. *The neuronal states and phenomenal states are both closely related to consciousness, but they exhibit completely different features. They were discovered using first-person and third-person approaches respectively. How to connect them together has become one of the important issues in the study of consciousness.*

5. Temporal Naturalization of First-person and Third-person Perspectives through Their Timescales

5.1. Temporal Naturalization of First and Third-Person Perspectives

So far we basically describe the methodological situation in the investigation of consciousness. First-person methods are used to describe and investigate the phenomenal features of consciousness while third-person approaches are employed when investigating its neuronal features. This reflects the methodological parallelism of first and third-person perspectives with respect to the neuronal and phenomenal features of consciousness. However, that methodological parallelism also affects our view of neuronal and phenomenal features of consciousness.

Due to such methodological parallelism, we remain unable to take into view the direct connection of neuronal and phenomenal states beyond their differences as states above. For instance, we remain unable to take into view their possibly shared features, that is, their "common currency". This makes necessary the development of a methodological strategy that, extending beyond the current first and third-person perspective parallelism, directly converges and integrates both perspectives. We will argue that such connecting features are timescales as both first and third-person perspectives can be characterized by them albeit in different degrees.

That, in turn, allows us to take into view the direct integration of neuronal and phenomenal states which, as we assume, consists in their temporal connection through timescales. Hence, we postulate direct convergence of the method of consciousness investigation with consciousness itself. Employing a particular methodological strategy will, through its temporal features, e.g., timescales, reveal the direct connection between the two states, e.g., phenomenal and neuronal, to which it is applied. We thus naturalize the first and third-person perspectives themselves. They are usually taken as a "given" without their underlying naturalistic substrate. This renders impossible to draw direct connection of first and third-person perspectives with the naturalistic features of consciousness including its peculiar connection of phenomenal and neuronal states. Research method, e.g., first- and third-person perspective, and research subject, e.g., relation of neuronal and phenomenal states, are here distinct from each other. That changes once one considers first and third-person perspectives by themselves in a naturalistic context as for instance in the temporal terms of their timescales. This can be understood as a form of temporal naturalization in a broad meaning of the term that does not entail reductionism or otherwise (Northoff, 2014; 2018).

More generally, such temporal naturalization of first and third-person perspective makes possible the convergence of research method, e.g., first and third-person perspective approaches, and research subject, e.g., neuronal and phenomenal features of consciousness. This is highly plausible given that our research methods, e.g., first and third-person perspective are also based and rely on the very same brain in which we aim to investigate the relation of neuronal and phenomenal features of consciousness. Hence, research method and research subject converge onto the same underlying feature, the brain, and are therefore both constrained by one and the same underlying features. We assume that these features of the brain constraining both research method, e.g., first and third-person perspective and research subject, e.g., neuronal and phenomenal features of consciousness, consist in the temporal features of the brain, that is, its timescales. That shall be elaborated in the following by showing distinct timescales of first and third-person perspectives in this section (Yue *et al.*, 2021). That, in turn, allows us to take into view the direct connection of neuronal and phenomenal features of consciousness through their temporal features, e.g., timescales, that shall be described in the next section.

One may contest that especially the first-person perspective with its grasp of phenomenal features can be naturalized in a temporal way. For instance, Thomas Nagel argues for a dichotomy between phenomenology and space-time. According to Nagel, the objective spatiotemporal order of the material world and the subjective phenomenological order of self-experience are different in principle: the former is objective and spatiotemporal, whereas the latter is subjective, phenomenological, and therefore non-spatiotemporal (Nagel, 2000). Nagel would thus reject our proposal of the temporal naturalization of the first-person perspectives and its phenomenal states as that would be to conflate or confuse the spatiotemporal features of the third-person perspective with the non-spatiotemporal features of the first-person perspective.

We argue that there is no such confusion as Nagel refers to a different meaning of space time than us. He seems to refer only to external time and space in the outer or external world while neglecting that the internal phenomenological world of experience also exhibits its own inner time space (Northoff *et al.*, 2020), an idea which can be traced to Kant's distinction of inner and outer time (Kant, 1998) and has been applied to the brain with its spontaneous activity (Northoff, 2012). Simply put, the phenomenal state of the first-person perspective has internal time and space (Mindt, 2021), while the objective world of the third-person perspective has external time and space. They are not principally different but characterized by different scales of space and time like different timescales. That shall be discussed in the following in more detail.

5.2. Longer Timescales of First-Person Perspectives

The first-person perspective operates at a relatively macroscopic spatiotemporal scale that derives from and is based on the nesting between different spatiotemporal scales. It is mainly expressed in longer timescales, that is, longer duration of its processing. William James, in his book Principles of Psychology, argued that consciousness is a continuous flow process, an ever-changing state of mind (James, 2007). James likened consciousness to a river that flows continuously and contains a myriad of perceptions, thoughts, emotions, and conscious content. He emphasized the continuity and uninterruptedness of consciousness as a continuous mental flow. The first-person perspective is the center of a continuous mental flow responsible for unifying and organizing the various elements of consciousness.

Therefore, the first-person point of view is based on sameness and is able to maintain a sense of continuity and stability in the midst of constant change and development. We know that various cells within the human body have different lifespans. For example, epidermal cells are usually renewed in about 2 to 4 weeks, red blood cells have a lifespan of about 120 days, intestinal lining cells are renewed in about a few days to a week, and brain cells are almost never renewed in human adulthood until the death of life. Although body parts are constantly changing on different timescales, the first-person perspective is able to nest and unify them, thus maintaining stability and continuity within the subject. This allows consciousness to span longer timescales, a day, a month, a year, or even a lifetime.

This view is also supported by neuroscientific evidence. Brain activity measures from fMRI and EEG for the same subjects, as well as psychological investigations from the first-person Consciousness Scale, show high correlations between resting-state scale-free activity and all psychological tasks exploring first-person specificity. Scale-free features of the resting state appear to play a central role in regulating different components of the subject's conscious specificity. The brain's long-duration activity on the scale-free has a strong shaping effect on first-person experience (Huang *et al.*, 2016; Wolff *et al.*, 2019).

The first-person perspective has a longer time-scale correlation. Long timescale correlations can link different timescales together through integrating and embedding them in a nested way (like the different Russian dolls). The first-person perspective places consciousness in the world through the timescales of the brain and constitutes a link between the world and the brain through their partially shared timescales. Moreover, the primary characteristic of the first-person perspective is its continuous change, e.g., its dynamic nature. The state of phenomena in the first-person perspective is always in flux. As many philosophers have pointed out, a specific internal feeling of ours displays a fleeting character. Hence, in the experience of reality, one phenomenal state will always be in the process of transitioning to the next.

5.3. Shorter Timescales of the Third-Person Perspectives

The third-person perspective operates at a relatively microscopic spatiotemporal scale that derives from and is based on the limitations of the inherent spatiotemporal scales. It is mainly expressed in shorter timescales, that is, in shorter duration. Compared to the direct knowability of the first-person perspective, the third-person perspective is always limited (Apperly and Carroll, 2009). It is precisely this limitation that breaks the inaccessibility of the third-person perspective – this is represented indirectly through specific forms such as language, behavior, artifacts, etc. Thereby allowing it to take on a certain visible character. Therefore, the third-person perspective is based on difference, and it always shows different states in a specific timescale. But at the same time, due to the existence of restrictions, it can only be within a fixed range.

The third-person perspective is limited to short timescale correlations with various specific shorter durations. For example, from the third-person perspective we learn that dogs can distinguish a wider variety of odors, bats can receive sound waves of a wider range of frequencies, and dragonflies can see a greater variety of colors. But for different subjects, these features cannot be linked, which makes the data obtained from the third-person perspective always limited. But assuming the existence of an animal that has excellent smell, hearing, and sight at the same time, it is conceivable that its first-person perspective would integrate on a longer timescale those features that are exhibited on a shorter timescale in different animals. Additionally, another main characteristic of the third-person perspective is that it is more static. The state of neurons in the third-person perspective is observed at the level of slices of different discrete points in time. For example, the firing of some cells under a certain moment in time, then the blood flow in some brain regions under a certain moment in time, and so on. Thus, neuronal states on short timescales can provide more precise data, but, being more static and less dynamic, cannot be directly linked to consciousness.

How can we understand the limitations of this third-person perspective? As is well known, it is difficult for people to achieve unity of thought with each other. Because thoughts are distributed dynamically over a longer time scale, and are only accessible to the first-person perspective. However, people can express their viewpoints through language, behavior, and artifacts, in order to achieve the goal of exchanging thoughts. They are the representational forms of the first-person thoughts from a third-person perspective, just as a shorter time scale partially overlaps with a longer time scale, a third-person viewpoint is a partial reflection of a first-person thought. It is precisely because of these limitations that the third-person perspective, unlike the first-person perspective, can reach a consensus between different subjects and thus be objectively visible on a shorter time scale.

Taken together, we define the first-person perspective by its longer timescales and the third-person scale by shorter timescales. Both timescales of first and third-person perspectives are important. If there is no third-person shorter timescale, then the nesting of the first-person longer timescale does not occur. If there is no first-person longer timescale, then the third-person shorter timescale is disconnected from consciousness and thus loses its meaning of existence. This reinterpretation is not only philosophically and scientifically sound, but can be compatible with either the first- or third-person approach. Simply put, the dynamic nature of the longer timescale of the first-person perspective inevitably leads to the qualitative, private, sensitive character of the first-person approach; the static nature of the shorter timescale of the third-person perspective inevitably leads to the quantitative, public, robust character of the third-person approach. We can now consider how to relate the two. Although the first-person perspective and the third-person perspective are based on different timescales, i.e., there are longer and shorter timescales. However, both are, after all, based on timescales or spatiotemporal relationships. Therefore, it is possible to combine and connect them in a temporal way.

6. Converging Research Method and Research Subject: Connecting Neuronal and Phenomenal Features of Consciousness through Their Timescales

6.1. Integrating First-Person Perspective and Third-Person Perspective Through Their Temporal Features

We are now ready to take into view how timescales allow for connecting first- and third-person perspectives. Despite their different timescales, e.g., short and long, first- and third-person perspective nevertheless share their characterization by timescales as such. This raises the question how and what features of the timescales make possible the connection, convergence and ultimately the integration of first and third-person perspectives in our investigation of the neuronal and phenomenal features of consciousness (Prencipe and Zelazo, 2005). Without elaborating the full details, we propose three temporal features of timescales that make possible to connect and converge first and third-person perspectives, namely coupling, matching and synchronization.

Coupling describes the interaction and mutual influence of first-person and third-person perspectives at different timescales (Choifer, 2018). Let's assume that longer timescales can interact with shorter timescales. This is because the longer timescale will always contain the shorter timescale. Take the phenomenon of boiling water as an example. Water molecules gain more kinetic energy due to heat, i.e., they vibrate and move rapidly, which causes the entire water liquid to start boiling and brings about the formation of bubbles and the release of steam. The change of kinetic energy of the water molecules at the microscopic scale is closely linked to the change of form of the water liquid at the macroscopic scale. This is the coupling between different timescales.

Matching describes the adaptation and interconnection of the different timescales of the first-person and third-person perspectives. Longer timescales can be linked to shorter timescales through intermediate timescales, which is a process of matching each other. For example, the world is a macro scale, the brain is a micro scale, and consciousness is an intermediate timescale. The world as an external being and the brain as a being in its own right are connected by the intermediary role of consciousness. In real life, the brain acquires information about the world through consciousness and is anchored in the world. The world relies on consciousness to acquire meaning from the brain and to be artificially transformed. Therefore, there is also a sort of intermediate timescale between the first-person perspective and the third-person perspective, which allows the two to be matched according to specific contexts, and in the process of matching they decide whether or not to adapt to each other. Take surfing for example. A person on a surfboard glides smoothly and quickly through the waves. The realization of this behavior requires the surfer to balance the relationship between the waves and the surfboard. The wave serves as the relative macro-scale and the surfboard serves as the relative micro-scale. The surfer acts as an intermediate and connecting timescale between the two and matches the different scales of the seawaves and the surfboard.

Synchronization describes the alignment and locking of first-person and third-person perspectives to each other at different timescales. Neuroscience research has found that during interaction between two or more subjects, the cortical activity between different individuals will show the phenomenon of synchronization, referred to as Interpersonal Neural Synchronization: it actually refers to the temporal alignment of neuronal signals between different individuals, which can reflect the correlation of different signals on a timescale (Czeszumski *et al.*, 2020; Lotter *et al.*, 2023). Interpersonal neural synchronization is a short timescale synchronization, and similar synchronization exists in long timescales. For example, different subjects all produce phenomenal states about heat when they see fire, and they all produce phenomenal states about cold when they see snow and ice. Although these phenomenal states may be subtly different, there is still a high degree of similarity. Therefore, the brains of different individuals share similar functional structures, and the qualia store of different individuals share similar kinds of states (Northoff, 2023). This provides the possibility of obtaining synchronization between the different timescales of the first-person perspective and those of the third-person perspective.

To summarize, first- and third-person approaches can be integrated through their different timescales. The longer time-scale of the first-person perspective and the shorter time-scale of the third-person perspective are integrated through coupling, matching, and synchronization. This enables the direct connection of first and third-person perspectives and consequently of neuronal and phenomenal states. Moreover, as suggested by the hypothesis of temporal features serving as common currency of neuronal and phenomenal states, the features that connect first and third-person perspectives, are also those that connect neuronal and phenomenal features, e.g., timescales. Hence, there is strong convergence of research methodology and research subject. This provides a methodological extension of the current neurophenomenological approach by directly linking first and third-person perspectives through their timescales. This new neurophilosophical approach provides us with the possibility of combining the neuronal state with the phenomenal state.

6.2. Connecting Neuronal States and Phenomenal States Extends Through Spatiotemporal Expansion

Based on a neurophilosophical approach with such more holistic perspective, we can finally consider the connection between neuronal states and phenomenal states. Here, spatiotemporal features serve as a common currency linking concepts and facts. To summarize, different spatiotemporal scales cross the boundaries of the brain, the body, and the world, giving rise to consciousness and forming a direct connection between neuronal and phenomenal states. Thus, consciousness is primarily conceived through the spatiotemporal approach proposed by the Temporospatial Theory of Consciousness (TTC). The TTC explains the correspondence between neuron activity and different dimensions of consciousness through four spatiotemporal mechanisms.

The details are as follows. Spatiotemporal nestedness: it organizes neuronal activity at different frequencies and locations within the brain so that functional areas within the brain are connected to each other, mirroring waking and normal states of consciousness; Spatiotemporal alignment: it processes and encodes multiple external inputs on the inner neural timescale of the brain, which determines the content selection and unified formal structure of consciousness; Spatiotemporal expansion, it extends the spatiotemporal characteristics of actual physical stimuli beyond their own degree, got content at the same time, and realizes the reproduction of the original stimulus by phenomenal consciousness; Spatiotemporal globalization, it recruits relevant functional regions and activity frequencies in the global scope of the brain, so that the content of consciousness can be accessed and controlled rationally at the cognitive level (Northoff, 2022).

According to TTC, spatiotemporal expansion is the use of the spontaneous activity of neurons to expand the spatiotemporal characteristics of actual physical stimuli beyond themselves. When a physical stimulus enters the brain in a specific spatiotemporal form and triggers induced neuronal activity, the spontaneous activity of the brain expands the spatiotemporal scale of that induced neuronal activity on its own spatiotemporal scale. That is, temporal persistence and spatial expansion, which in turn produces overflow based on the content of consciousness. That is, the longer timescale of the first-person perspective is able to extend the shorter timescale of the third-person perspective in a virtual way which in turn leads to the transformation of neuronal states into phenomenal states.

Specifically, we used third-person brain imaging techniques to search for physiological markers of neuronal composition on the one hand to obtain objective data at shorter spatiotemporal scales, and first-person introspective and phenomenological approaches to clarify the phenomenal state of the experience on the other hand to obtain subjective reports at longer

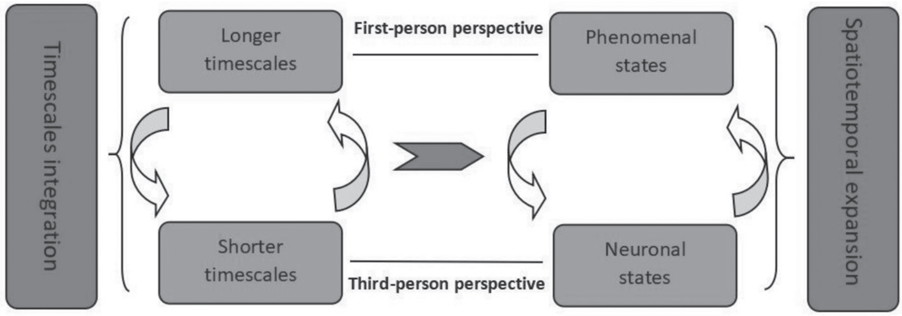

FIG. 2. Schematic illustration of the connection neuronal states and phenomenal states. *The integration of different timescales, e.g.; shorter and longer durations, leads to the connection between different perspectives, e.g., first- and third-person perspective, which enables one to connect neuronal and phenomenal states. Thus, consciousness is described as the former transforming into the latter through spatiotemporal expansion.*

spatiotemporal scales. The results based on the different spatiotemporal scales are then constrained in an attempt to establish a stable link between the different perspectives. Finally, a holistic perspective-based neurophilosophical approach is utilized to integrate the different timescales into a unified context. And in this context the transitions between neuronal and phenomenal states are described. This transition is based on coupling, matching and synchronization between different timescales (Piper, 2019), establishing a link between neuronal and phenomenal states.

7. Conclusion

How can we take into view the connection of neuronal and phenomenal states of consciousness like their shared features? We currently do not know. While neurophenomenology provides methodological tools for investigating both neuronal and phenomenal features, it still suffers from methodological parallelism of first and third-person perspectives. We here suggest that first and third-person perspectives can be naturalized themselves in temporal terms, that is, by their distinct timescales. Though exhibiting distinct timescales, first and third-person perspectives nevertheless share one common feature, namely timescales as such. This, as we argue, makes possible to connect, converge and integrate them thus overcoming the methodological parallelism of current neurophenomenology.

Such temporal integration of first and third-person perspectives carries major reverberation for our search for the connection of neuronal and phenomenal states of consciousness. If first and third-person perspectives are linked through their timescales, one would assume that the neuronal and phenomenal features of consciousness are also connected by their timescales. This is plausible to assume given that both consciousness and perspectives, e.g., research subject and research method, share one and the same underlying basis, namely the brain and its various different timescales. They must thus be constrained by the same features that also characterize the brain; for that, we assume that timescales provide a strong candidate. Both our research method and research subject may then be traced to one and the same feature that also characterizes the brain, that is, timescales.

In a nutshell, we propose that timescales allow converging first and third-person perspectives which, in turn, allows us to take into view the intrinsically temporal connection of neuronal and phenomenal states of consciousness through the brain's timescales. This aligns well with a primarily temporal characterization of consciousness which has been suggested recently (Kent and Wittmann, 2021), and elaborated in full detail in the TTC (Northoff and Huang 2017; Northoff and Zilio, 2022).

References

Aagaard P. (2003). Training-induced changes in neural function. *Exercise and Sport Sciences Reviews*, 31, 2: 61-67, doi: 10.1097/00003677-200304000-00002.

Apperly I.A. and Carroll D.J. (2009). How do symbols affect 3-to 4-year-olds' executive function? Evidence from a reverse-contingency task. *Developmental Science*, 12, 6: 1070-1082, doi: 10.1111/j.1467-7687.2009.00856.x.

Aydede M. (2019). Is the experience of pain transparent? Introspecting phenomenal qualities. *Synthese*, 196: 677-708.

Block N. (1981). The Mind-Body Problem. *Scientific American*, 244, 1: 114-123.

Block N. (1995). On a confusion about a function of consciousness. *Behavioral and Brain Sciences*, 18, 2: 227-247, doi: 10.1017/s0140525x00038188.

Botvinick M. and Braver T. (2015). Motivation and cognitive control: From behavior to neural mechanism. *Annual Review of Psychology*, 66: 83-113, doi: 10.1146/annurev-psych-010814-015044.

Brentano F. (1995) [1874[1]]. *Psychology from an Empirical Standpoint*. London: Routledge and Kegan Paul.

Chalmers D.J. (1995). Facing up to the problem of consciousness. *Journal of Consciousness Studies*, 2, 3: 200-219, doi: 10.1093/acprof:oso/9780195311105.003.0001.

Choifer A. (2018). A New Understanding of the first-person and third-person Perspectives. *Philosophical Papers*, 47, 3: 333-371, doi: 10.1080/05568641.2018.1450160.

Crick F. and Koch C. (1990). Towards a neurobiological theory of consciousness. *Seminars in the Neurosciences*, 2: 263-275.

Cuthbertson L.M., Robb Y.A. and Blair S. (2020). Theory and application of research principles and philosophical underpinning for a study utilising interpretative phenomenological analysis. *Radiography*, 26, 2: 94-102, doi: 10.1016/j.radi.2019.11.092.

Czeszumski A., Eustergerling S., Lang A., Menrath D., Gerstenberger M., Schuberth S. and König P. (2020). Hyperscanning: a valid method to study neural inter-brain underpinnings of social interaction. *Frontiers in Human Neuroscience*, 14: 39, doi: 10.31219/osf.io/7ey4w.

Dehaene S., Charles L., King J.R. and Marti S. (2014). Toward a computational theory of conscious processing. *Current Opinion in Neurobiology*, 25: 76-84, doi: 10.1016/j.conb.2013.12.005.

Donahoe J.W. (2002). Behavior analysis and neuroscience. *Behavioural Processes*, 57, 2-3: 241-259, doi: 10.1016/s0376-6357(02)00017-7.

Dunbar R.I. and Shultz S. (2007). Understanding primate brain evolution. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 362, 1480: 649-658, doi: 10.1098/rstb.2006.2001.

Edelman G.M. (1993). Neural Darwinism: selection and reentrant signaling in higher brain function. *Neuron*, 10, 2: 115-125, doi: 10.1016/0896-6273(93)90304-a.

Flinker A., Korzeniewska A., Shestyuk A.Y., Franaszczuk P.J., Dronkers N.F., Knight R.T. and Crone N.E. (2015). Redefining the role of Broca's area in speech. *Proceedings of the National Academy of Sciences*, 112, 9: 2871-2875, doi: 10.1073/pnas.1414491112.

Fodor J.A. (1981). The Mind-Body Problem. *Scientific American*, 244, 1: 114-123, doi: 10.1038/scientificamerican0181-114.

Fossey E., Harvey C., McDermott F. and Davidson L. (2002). Understanding and evaluating qualitative research. *Australian & New Zealand Journal of Psychiatry*, 36, 6: 717-732, doi: 10.1046/j.1440-1614.2002.01100.x.

Ge S., Liu H., Lin P., Gao J., Xiao C. and Li Z. (2018). Neural basis of action observation and understanding from first- and third-person perspectives: An fMRI study. *Frontiers in Behavioral Neuroscience*, 12: 283, doi: 10.3389/fnbeh.2018.00283.

Huang Z., Obara N., Davis H.H.IV., Pokorny J. and Northoff G. (2016). The temporal structure of resting-state brain activity in the medial prefrontal cortex predicts self-consciousness. *Neuropsychologia*, 82: 161-170, doi: 10.1016/j.neuropsychologia.2016.01.025.

Jack A.I. and Shallice T. (2001). Introspective physicalism as an approach to the science of consciousness. *Cognition*, 79, 1-2: 161-196, doi: 10.1016/s0010-0277(00)00128-1.

James W. (2007). *The Principles of Psycholog*, vol. I. New York: Cosimo, Inc.

Kant I. (1998) [1781]. *Critique of Pure Reason*. In: *Cambridge Edition of the Works of Immanuel Kant*, ed. by P. Guyer and A. Wood. Cambridge: Cambridge University Press. (Ed or. 1781/1787 Hartnoch, Riga).

Kent L. and Wittmann M. (2021). Time consciousness: The missing link in theories of consciousness. *Neuroscience of Consciousness*, 2021, 2, niab011, https://doi.org/10.1093/nc/niab011.

Köhler W. (1967). Gestalt psychology. *Psychologische Forschung*, 31, 1: xviii-xxx, doi: 10.1007/bf00422382.

Lamme V.A and Roelfsema P.R. (2000). The distinct modes of vision offered by feedforward and recurrent processing. *Trends in Neurosciences*, 23, 11: 571-579, doi: 10.1016/s0166-2236(00)01657-x.

Levine J. (1983). Materialism and qualia: The explanatory gap. *Pacific Philosophical Quarterly*, 64, 4: 354-361, doi: 10.1111/j.1468-0114.1983.tb00207.x.

Levine J. (2006). Eight phenomenal concepts and the materialist constraint. In Alter T. and Walter S., eds. *Phenomenal Concepts and Phenomenal Knowledge: New Essays on Consciousness and Physicalism*. Oxford: Oxford University Press: 145-166.

Lotter L.D., Kohl S.H., Gerloff C., Bell L., Niephaus A., Kruppa J.A. and Konrad K. (2023). Revealing the neurobiology underlying interpersonal neural synchronization with multimodal data fusion. *Neuroscience & Biobehavioral Reviews*, 146, 105042, doi: 10.1016/j.neubiorev.2023.105042.

Metag J. (2021). Tension between visibility and invisibility: Science communication in new information environments. *Studies in Communication Sciences*, 21, 1: 129-144, doi: 10.24434/j.scoms.2021.01.009.

Mindt G. (2021). Not all structure and dynamics are equal. *Entropy*, 23, 9: 1226, doi: 10.3390/e23091226.

Nagel T. (1974). What is it like to be a bat? *The Philosophical Review*, 83, 4: 435-450.

Nagel T. (2000). The psychophysical nexus. In Boghossian P. and Peacocke C., eds. *New essays on the a priori*. Oxford: Oxford University Press: 433-471.

Northoff G. (2004). *Philosophy of the Brain*. Amsterdam: John Benjamins Publishing.

Northoff G. (2012). Immanuel Kant's mind and the brain's resting state. *Trends in Cognitive Sciences*, 16, 7: 356-359, doi: 10.1016/j.tics.2012.06.001.

Northoff G. (2014). *Unlocking the Brain II*. Oxford: Oxford University Press.

Northoff G. (2022a). From shorter to longer timescales: Converging Integrated Information Theory (IIT) with the Temporo-Spatial Theory of Consciousness (TTC). *Entropy*, 24, 270, doi: 10.3390/e24020270.

Northoff G. (2022b). Temporo-spatial Theory of Consciousness (TTC) – Bridging the gap of neuronal activity and phenomenal states. *Behavioural Brain Research*, 424, 113788.

Northoff G. (2022c). Non-reductive neurophilosophy – What is it and how it can contribute to philosophy. *Journal of NeuroPhilosophy*, 1, 1: 17-30, doi: 10.1007/978-3-030-95369-0_5.

Northoff G. (2023). *Neuropsychoanalysis: A Contemporary Introduction*. London: Routledge.

Northoff G. (2024). Is Heidegger's fundamental ontology really fundamental? From anthropological to ecological fundamental ontology. *The Humanist* (forthcoming)

Northoff G. and Huang Z. (2017). How do the brain's time and space mediate consciousness and its different dimensions? Temporo-spatial theory of consciousness (TTC). *Neuroscience & Biobehavioral Reviews*, 80: 630-645, doi: 10.1016/j.neubiorev.2017.07.013.

Northoff G. and Scalabrini A. (2021). "Project for a spatiotemporal neuroscience" – brain and psyche share their topography and dynamic. *Frontiers in Psychology*, 12, doi: 10.3389/fpsyg.2021.717402.

Northoff G., Wainio-Theberge S. and Evers K. (2020a). Is temporo-spatial dynamics the "common currency" of brain and mind? In quest of "spatiotemporal neuroscience". *Physics of Life Reviews*, 33: 34-54, doi: 10.1016/j.plrev.2019.05.002.

Northoff G., Wainio-Theberge S. and Evers K. (2020b). Spatiotemporal neuroscience – what is it and why we need it. *Physics of Life Reviews*, 33: 78-87, doi: 10.1016/j.plrev.2020.06.005.

Overgaard S. (2006). The problem of other minds: Wittgenstein's phenomenological perspective. *Phenomenology and the Cognitive Sciences*, 5: 53-73, doi: 10.1007/s11097-005-9014-7.

Piper M.S. (2019). Neurodynamics of time consciousness: An extensionalist explanation of apparent motion and the specious present via reentrant oscillatory multiplexing. *Consciousness and Cognition*, 73, 102751.

Prencipe A. and Zelazo P.D. (2005). Development of affective decision making for self and other: Evidence for the integration of first- and third-person perspectives. *Psychological Science*, 16, 7: 501-505, doi: 10.1111/j.0956-7976.2005.01564.x.

Pringle J., Drummond J., McLafferty E. and Hendry C. (2011). Interpretative phenomenological analysis: A discussion and critique. *Nurse Researcher*, 18, 3, doi: 10.7748/nr2011.04.18.3.20.c8459.

Rizzuto A.M. (1993). First person personal pronouns and their psychic referents. *The International Journal of Psycho-Analysis*, 74, 3, 535-546.

Schneider S. and Velmans M., eds. (2017). *The Blackwell Companion to Consciousness*. Oxford: John Wiley & Sons.

Sedlmeier P., Eberth J., Schwarz M., Zimmermann D., Haarig F., Jaeger S. and Kunze S. (2012). The psychological effects of meditation: a meta-analysis. *Psychological Bulletin*, 138, 6: 1139-1171, doi: doi.org/10.1037/a0028168.

Shapiro L. (2011). Descartes's pineal gland reconsidered. *Midwest Studies in Philosophy*, 35, 1: 259-286.

Snyder A.Z. and Raichle M.E. (2012). A brief history of the resting state: the Washington University perspective. *NeuroImage*, 62, 2: 902-910, doi: 10.1016/j.neuroimage.2012.01.044.

Storm J.F., Boly M., Casali A.G., Massimini M., Olcese U., Pennartz C.M. and Wilke M. (2017). Consciousness regained: disentangling mechanisms, brain systems, and behavioural responses. *Journal of Neuroscience*, 37, 45: 10882-10893, doi: 10.1523/JNEUROSCI.1838-17.2017.

Sun Y., Pan Z. and Shen L. (2008). Understanding the third-person perception: Evidence from a meta-analysis. *Journal of Communication*, 58, 2: 280-300, doi: 10.1111/j.1460-2466.2008.00385.x.

Symons J. (2008). Intuition and philosophical methodology. *Axiomathes*, 18: 67-89.

Thompson E. (2014). *Waking, Dreaming, Being: Self and Consciousness in Neuroscience, Meditation, and Philosophy*. New York: Columbia University Press.

Tune G.S. (1964). Response preferences: A review of some relevant literature. *Psychological Bulletin*, 61, 4: 286-302, doi: 10.1037/h0048618.

Uttar W.R. (2011). *Mind and Brain: A Critical Appraisal of Cognitive Neuroscience*. Cambridge: MIT Press.

Whitehead A.N. (1925). *Science and the Modern World*. London: Macmillan.

Wolf A., Di Giovanni D.A., Gomez-Pilar J., Nakao T., Huang Z., Longtin A. and Northoff G. (2019). The temporal signature of self: Temporal measures of resting-state EEG predict self-consciousness. *Human Brain Mapping*, 40, 3: 789-803, doi: 10.1002/hbm.24412.

Yue C., Long Y., Ni C., Peng C. and Yue T. (2021). Valence of temporal self-appraisals: A comparison between first-person perspective and third-person perspective. *Frontiers in Psychology*, 12, doi: 10.3389/fpsyg.2021.778532.

Zhang J. and Northoff G. (2022). Beyond noise to function: Reframing the global brain activity and its dynamic topography. *Communications Biology*, 5, 1, 1350, doi: 10.1038/s42003-022-04297-6.